ReCare AI

Explainable & Fair Hospital Readmission Risk Prediction

ReCare AI estimates the risk of **30-day hospital readmission** to support discharge planning and follow-up prioritization.

⚠ Prototype decision-support tool. This risk estimate is not clinically validated and does not replace professional medical judgment.

1. Patient Information

Race: Caucasian

Gender: Female

Age Group: [50-60)

2. Hospitalization Information

Time in Hospital (days): 4

Number of Lab Procedures: 40

Number of Procedures: 1

Number of Medications: 16

Number of Diagnoses: 7

Admission Type: 1

3. Previous Healthcare Utilization

Previous Outpatient Visits: 0

Previous Emergency Visits: 0

Previous Inpatient Visits: 0

4. Diabetes Management

Diabetes Medication Prescribed

Insulin Treatment

Metformin Treatment

Yes
HbA1c Test Result

No
Blood Glucose Test Result

No

None

None

5. Readmission Risk

🔍 Predict Readmission Risk

Prediction completed successfully!

Estimated 30-Day Readmission Risk

38.7%

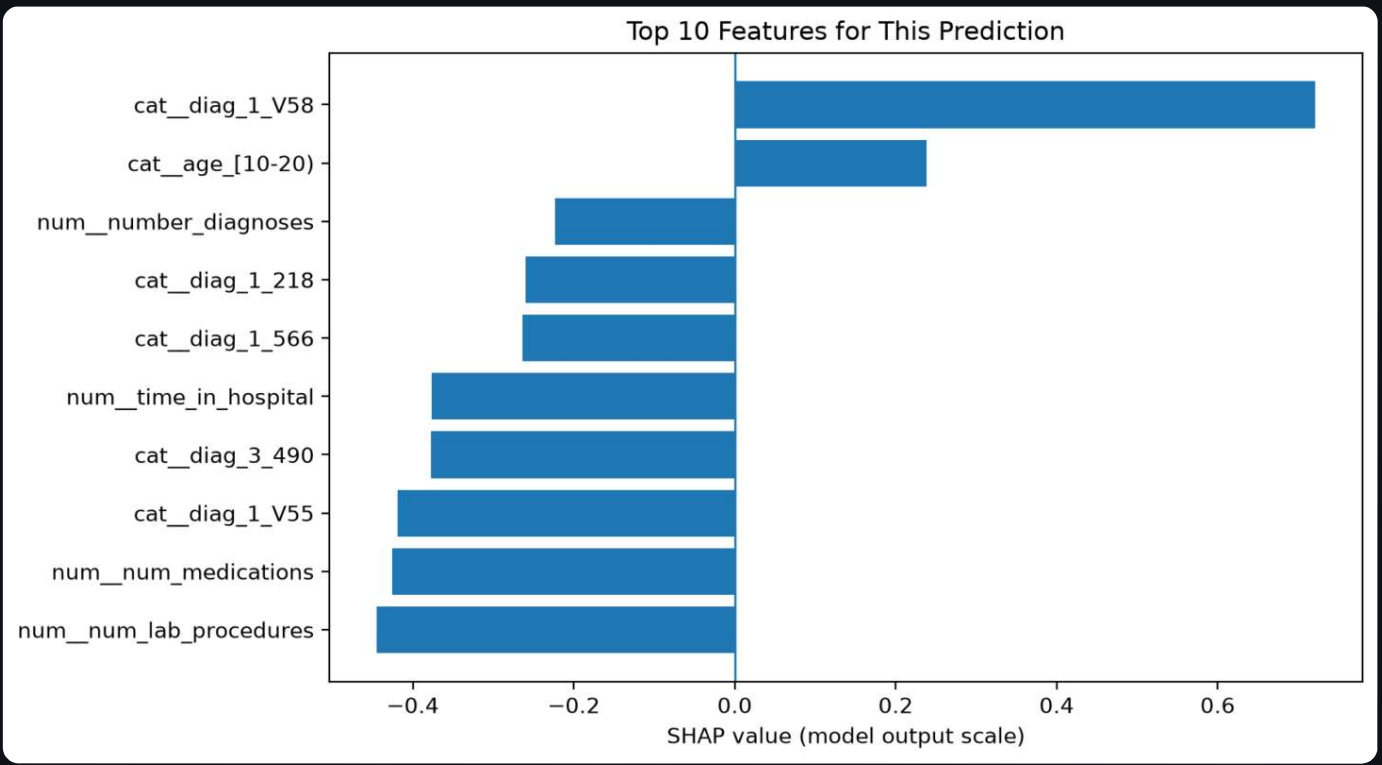
Risk Category

MODERATE RISK

Follow-up Priority

🟠 MODERATE PRIORITY — Consider additional follow-up and discharge-planning review.

6. SHAP Explanation for This Patient



› View patient SHAP values

Positive SHAP values push the explained model output higher; negative values push it lower. For TreeExplainer, values are often on the raw model-output (log-odds) scale.

7. Global SHAP Analysis

Global SHAP summarizes influential model features across a sample of the dataset. It is different from the explanation for one patient.

📊 Calculate Global SHAP Analysis

ReCare AI is an educational prototype, not a clinically validated medical device. SHAP explains model behavior and does not establish causation, fairness, or clinical validity.